

Stick Game

Time limit: 1.00 s

Memory limit: 512 MB

Consider a game where two players remove sticks from a heap. The players move alternately, and the player who removes the last stick wins the game.

A set $P = \{p_1, p_2, \dots, p_k\}$ determines the allowed moves. For example, if $P = \{1, 3, 4\}$, a player may remove 1, 3 or 4 sticks.

Your task is find out for each number of sticks $1, 2, \dots, n$ if the first player has a winning or losing position.

Input

The first input line has two integers n and k : the number of sticks and moves.

The next line has k integers $p_1, p_2, \dots, p_k$ that describe the allowed moves. All integers are distinct, and one of them is 1.

Output

Print a string containing n characters: W means a winning position, and L means a losing position.

Constraints

- $1 \leq n \leq 10^6$
- $1 \leq k \leq 100$
- $1 \leq p_i \leq n$

Example

Input	Output
10 3 1 3 4	WLWWWLWLW